

Methods of Protection

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Organizations implement multiple protection mechanisms to secure information systems.

Methods of Protection:

Authentication

Authorization

Encryption

Firewalls

Antivirus Software

Backup and Recovery

Access Control

Security Policies

Physical Security

User Awareness Training

Authentication

Definition

Authentication verifies the identity of a user before granting access to a system.

Purpose

Ensures only legitimate users can access resources.

Examples

Username and Password

Biometric Authentication

One Time Password (OTP)

Multi Factor Authentication (MFA)

Authorization

Definition

Authorization determines what resources and operations a user is allowed to access after authentication.

Purpose

Prevents unauthorized access to sensitive information.

Example

Students can view examination results but cannot modify examination records.

Types

Role Based Access Control

Attribute Based Access Control

Rule Based Access Control

Encryption

Definition

Encryption converts readable information into an unreadable format to protect data from unauthorized access.

Benefits

Protects sensitive information

Secures online transactions

Maintains confidentiality

Example

Credit card information is encrypted during online banking and shopping transactions.

Encryption Process

Plain Text



Encryption Algorithm



Cipher Text



Decryption



Plain Text

Explanation

Only authorized users with the correct key can decrypt and read the information.

Firewalls

Definition

A firewall is a security device that monitors and controls incoming and outgoing network traffic.

Functions

Block unauthorized access

Filter malicious traffic

Protect internal networks

Monitor network activities

Firewall Diagram

Internet



Firewall



Internal Network

Explanation

The firewall acts as a protective barrier between trusted and untrusted networks.

Antivirus Software

Definition

Antivirus software detects, prevents, and removes malicious software from computer systems.

Functions

Virus Detection

Malware Removal

Real Time Protection

Scheduled Security Scans

Automatic Updates

Backup and Recovery

Definition

Backup and recovery mechanisms ensure that data can be restored after accidental deletion, hardware failure, or cyber attacks.

Benefits

Data Protection

Business Continuity

Disaster Recovery

Reduced Downtime

Types of Backup

Full Backup

Copies all data.

Incremental Backup

Copies only changes made since the last backup.

Differential Backup

Copies changes made since the last full backup.

Access Control

Definition

Access control restricts access to information and resources to authorized users only.

Methods

Passwords

Smart Cards

Biometric Systems

Security Tokens

Role Based Permissions

Security Policies

Definition

Security policies are formal guidelines that define security practices and responsibilities within an organization.

Examples

Password Policy

Data Handling Policy

Incident Response Policy

Acceptable Use Policy

Remote Access Policy

Physical Security

Physical security protects buildings, equipment, and other physical assets from unauthorized access and damage.

Examples

CCTV Cameras

Security Guards

Biometric Entry Systems

Access Control Doors

Visitor Management Systems

User Awareness Training

Definition

User awareness training educates employees and users about security risks and safe practices.

Importance

Reduces human errors

Improves security awareness

Prevents social engineering attacks

Strengthens organizational security

Topics Covered in User Awareness Training

Phishing Attacks

Password Management

Safe Internet Usage

Email Security

Social Engineering Awareness

Mobile Device Security

Data Privacy Protection

Importance of Protection Methods

Protect confidential information

Prevent unauthorized access

Ensure data integrity

Maintain service availability

Reduce security risks

Support business continuity

Summary

Authentication verifies user identity.

Authorization controls user permissions.

Encryption protects sensitive data.

Firewalls and antivirus software defend systems.

Backup and recovery ensure data availability.

Security policies and user training strengthen overall security posture.

Thank you

Any Questions?